

STATghost

Quick manual

clipboard → local engines → text/plot · v0.4.27 · 2026

A sniper, not a Swiss army knife. No editor; clipboard + R / Python / Julia.

1 30-second start

1. Unzip so `statghost.exe`, `statghost_rhost.exe`, `lexer/`, and `library/` stay together (rhost only for R.dll; `library/` for Explorer companion). Linux gtk2: unpack the `.tar.gz` and run `./statghost` (same sidecars: `lexer/`, `library/`, `statghost_py_repl.py`, `statghost_jl_repl.jl`).
2. Run `statghost.exe` (toolbar, top-right).
3. Under **Set** → **Session**, pick language **R**, **Python**, or **Julia** and (if needed) the engine path.
4. Click **Idle** → **Armed** (amber).
5. In another app, select code and copy (**Ctrl+C**) — or use the optional Capture hotkey with no selection (see below).
6. See results in **Console** / **Plot**.

2 Traffic lights

- **Set** — preferences (Appearance includes toolbar icon size 16 px / 24 px / 32 px, default 24).
- **Idle/Armed** — clipboard sniper on/off.
- **Kill** — stop *this app's* current engine child (see below).
- **Panel** — show/hide the output window (**Separados** float bar only; not duplicated on the Unidos tab strip).
- **Clear** — clear console and plot.
- **Unite/Split** — Separados ↔ Unidos (rightmost on the panel tab bar; switch glyph).
- **Background-progress gauge** — a slim hairline strip along the toolbar's bottom edge, always visible from the main window regardless of which dialog is open. Idle = fully hidden; lights up (determinate % or an indeterminate sweep) during CRAN catalog / mirror refresh, Install, or Update, then briefly flashes "done" before auto-collapsing. Text status stays on each control's own hint, same as before — the gauge is a second always-on-screen signal, not a replacement.

STATghost icon: menu *Help* / *About* / *OS global hotkeys* / *Quit* (click the icon). Drag the bar by the background or the name. Status shows engine badge and HK: **on** | **off**. Chrome glyphs are flat monochrome, tinted for the active theme.

3 Engines (R / Python / Julia)

Under **Set** → **Session** (Path | Options):

- **Use** — **R** | **Python** | **Julia**.
- **R**: home (empty = newest on this PC); Use `{Rterm.exe | R.dll}`. R.dll talks via sidecar `statghost_rhost.exe` (GUI never loads R.dll). **CRAN mirror (repos)** — text field, default `https://cloud.r-project.org`; sent as `options(repos=)` at session start and passed explicitly to every `install.packages()` call,

so a session is never left on R's own unset "@CRAN@" sentinel. Applies live to an already-Armed session. Also settable from Explorer's **Choose CRAN mirror** picker (see Explorer). **Update packages that need compilation (requires Rtools)** — checkbox, off by default. Some CRAN packages, when their Windows binary lags the just-published source release, can only be updated by compiling from source, which needs Rtools installed. Leave this off (the safe default) to skip such packages instead of risking a failed compile on a machine without Rtools; a hint under the checkbox reports whether an Rtools install was detected on this PC. Turn it on only if Rtools is installed.

- **Python:** interpreter path (empty = discover); Options — show Matplotlib window; prefetch imports on session start.
- **Julia:** path (empty = discover / juliaup); Options — show GR window when relevant. Plot panel PNGs use the same high-res size class as R/Python (Plots.jl `savefig` of `current()`).
- **R Options:** show `rgl` window when relevant.

Status badge examples: R.dll · 4.5.0, Python · 3.12, Julia · 1.11, or idle.

4 Kill

- Ends only STATghost's current child — `Rterm.exe`, `statghost_rhost.exe`, Python, or Julia — not RStudio / other IDEs.
- Works mid-eval (hard terminate). Toolbar returns to green Idle; click **Arm** for a fresh session.
- Changing language / paths / R Use while running kills the old session first.

5 Clipboard and dedup

When **Armed**, eval runs only if clipboard text changes. Copying the same selection again does **not** re-run (feature, not a bug). To repeat: change one character, or use a dedicated Capture hotkey (if set).

Current line without selection

Under **Set** → **Hotkeys** → **Capture**, record an **alternative** global chord (optional). That hotkey asks the foreground app to **Copy**, then STATghost evaluates the clipboard. In many editors (VS Code, Notepad++, RStudio, CudaText, ...), Copy with **no selection** copies the **current line** — so you can step a script line by line without selecting. Behaviour depends on the source app; if Copy does nothing without a selection, select text as usual (or use **Ctrl+C** when Armed). An optional CudaText plugin (VP-EB-1 / EB-1b) lives in the public companion <https://github.com/jcfaria/statghost-cudatext> (plugins/cuda_statghost). Copy that folder into CudaText py/ (or run `bash plugins/install_lab.sh` in the clone) and restart CudaText. It sends the current **selection** or, with an empty selection, the complete **statement** at the caret (brackets, operators, unbraced R `if` body) into the Armed clipboard path. Toolbar + side tab: Config, Arm/Idle, Start/Quit, Send, Source, Clear. **Source** writes the buffer to the shared TEMP slot `.paths[4]` (STATghostcom, TinnRcom pattern) and sends `source(.paths[4], echo=..., spaced=FALSE, encoding=...)` — no absolute editor path on the Console. Optional Config: join editor wraps into one Console line (when off, STATghost reprints the sent breaks and indent with `> / +`); Source echo/encoding. STATghost stays the receiver (one interactive instance, D50). After a send, the editor caret advances to the next code line.

Operational notices go to the console meta stream; the Console|Plot status strip sits at the **bottom of the panel** (engine badge, tips, and Update gauge when a package batch is running). After each eval, Console scroll can sit the new chunk at **Middle** (default), Top, or Bottom (Set → Behavior).

6 Panel

- **Console** — read-only **ATSynEdit** with Cuda lexers from the **lexer/** folder beside the EXE (R / Python / Julia; packs come from the companion **statghost-cudatext** at build time). R / Python / Julia use the STATghost Notes markers **#.** / **#..** / **#...** (R also has Programing / Datasets in the slim pack). Mixed text+plot samples use **#.** --- **GRAPHIC OUTPUT** --- as the two-stage marker (Note1 colour; TF also accepts the classic **#** --- form). **Ctrl+Tab** / **Shift+Ctrl+Tab** switch Console ↔ Plot. The active tab highlight follows the page after eval focus (Auto / Console / Plot). Echo prompts: R **>/+**, Python **>>>/ ...**, Julia **julia>/julia|** (continuation is classroom-style). Under the Console: small tweaks only inside an echoed **input** chunk (prompt locked; not a free editor), then **Ctrl+Enter** to re-eval (Settings → Shortcuts; Capture stays **Ctrl+C** when Armed). **C+Up/C+Down** (hold letter C, not Ctrl) jump previous/next input cell.
- **Chrome (Separados | Unidos)** — **Unite/Split** button on the tab bar (rightmost). **Separados**: floating traffic bar + panel; lights only on the float (panel shows the mode toggle). **Form único** / Unidos (default): one window — strip on the tab bar (brand | Set | Arm | Kill | Clear); floating bar hidden while the panel is open. Horizontal toolbar magnet: shared left edge under the bar; vertical: tops aligned, rail to the left of the panel.
- **Always on top** — Main traffic bar and Panel share one toggle (default on; chrome / toolbar). Uses Win32 topmost; Explorer has its own Top/Norm control.
- After eval, focus is **Auto** by default: **Plot** if this eval drew graphics, else **Console** (Set → Behavior to force Console, Plot, or return to source).
- **Plot** — high-res PNG (R, Python Matplotlib, Julia Plots.jl); zoom, history ←/→, erase-history control after the N/M counter, optional PDF (R). Set → Plot panel → Display chooses **Lazarus TImage** (default) or **ATSynEdit Gaps** (same AT family as Console); switch anytime.
- **Size** (Plot panel) — match plot to pane width (default on); PNG DPI slider (soft green → dark blue, default 288) shared by TImage and AT Gaps.
- Bookmarks (if enabled): **Ctrl+Shift+0..9** / **Ctrl+0..9**.

7 Window placement

Main, Panel, and Explorer store screen bounds in **window_geometry.json** (separate from prefs, same config folder — **Portable** <exe>/data/ or **AppData** %LOCALAPPDATA%/STATghost; see Config). Positions update live on move/resize (short debounce) and restore across sessions, including a float Explorer on a secondary monitor. Stale **(0, 0)** / unset **(-1, -1)** captures are ignored. On Linux (gtk2), restore clamps to the monitor **bounds** when the reported work area is not a small taskbar inset — a rotated/portrait second monitor keeps the saved vertical placement (bottom half stays bottom half).

8 Explorer (R)

Object browser for the active R session (VP-EX-1). Requires **library/STATghostcom_R/STATghostcom** beside the EXE (shipped in the zip) or the installed companion under AppData.

- **Open / hide**: traffic-bar **Expl** light, Unidos chrome button, or **Ctrl+E** (configurable under Set → Hotkeys).
- **Dock**: classroom default **Inside** (split the Console tab only; Plot full-width) with **L|R** relative to the editor (**explorer_inside_side** / **explorer_inside_width**; new-install default width 210 px, min 160). Toolbar: **Inside** is its own button; **Placement** is one parent for Left / Right / External (float). **Top/Norm** stay-on-top apply to External/Float only (Inside never topmost; default Top); **Auto/Fill** height (**Fill** keeps the Explorer status bar inside the work area); window size and Environment pane height are remembered (default eight compact package rows).
- **Environment**: **search()** list with **.GlobalEnv** always first. Click a row to list Objects in that environ-

ment. Icon actions: Mirror, Install, Update, Load, **Detach** (`detach( ... , unload=TRUE)`; always-visible option in Settings → Behavior), **Info** (`packageDescription` memo). **Hid/All** toggles hidden names (`all.names`).

- **Install/Update packages** (RStudio-style): **Install** opens a typeahead search over a local CRAN metadata cache (name + real **Title**, autocomplete as you type). **Update** reuses the same typeahead + zebra StringGrid family, scoped to already-installed packages with a newer cached CRAN version — columns Package | Installed | Available | Title, footer Total / Selected / Filter / Filter (%), All/None, and OK/Cancel (Cancel is the Enter/Esc default; double-click toggles a row check without applying the batch). Opening Update resyncs the installed side of the cache from the live library so packages removed outside STATghost do not linger. Update has no **Refresh index** (Install still owns catalog refresh). Metadata lives in an embedded SQLite3 cache (`packages.sqlite3`, same config folder as prefs) and only refreshes on demand (**Refresh index** in the Install dialog), never automatically at startup. **Refresh index** runs the CRAN fetch in a **detached Rscript** process (via `tools::CRAN_package_db()`, CRAN-only — never fans out to other configured repos) while the Install status bar shows “Indexing...”; the active Armed R session is never used for the download and is never Kill’d if a mirror is slow. Info/Load/Install/Update dialogs always open next to the real Explorer window and its Packages toolbar button — correct for every dock mode (Inside/Left/Right/Float), never collapsing onto the small sniper toolbar. Load/Install/Update lists render as a native grid with a plain bold dark header row (theme-independent by design — readable against the fixed-row background under every colour theme) and an optional zebra (alternate-row) banding mode, toggled under Set → Appearance (see Settings).
- **Choose CRAN mirror**: toolbar button beside the package icons opens a picker listing every mirror from R’s own `utils::getCRANmirrors()` (Name/Country/City/URL), with the recommended 0-Cloud automatic-CDN mirror always pinned at the top regardless of the filter text, plus “Open in browser”. The list is fetched by a detached **Rscript** (never the interactive session) into a small SQLite cache refreshed every 30 days or on demand; picking a mirror updates the same CRAN-mirror field under Set → Session → R (see Engines) and applies it live if a session is Armed.
- **Objects**: exclusive **Tree | Grid** (Name / Type / Class / Dim; Name fills remaining width). Grid **Type / Class** show short acronyms (`data.frame`→DF, `vector`→Vec, `function`→Fn, `matrix`→Mat, `array`→Arr, `list`→Lst, `table`→Tab, `other`→Oth; class tokens such as `integer`→Int, `numeric`→Num — unknown names stay as-is). Hover a row (or the Explorer status strip) for the full engine **type / class**. Filter by structural type (not S3/S4; icons match Tinn-R order) and optional Perl-compatible name regex (empty box; hint on hover; Enter or leave field stores MRU as `history_explorer_name_filter.json` under AppData STATghost; Up/Down recall). When Inside is narrow, eraser / Tree|Grid / Clear wrap to a **second** row under the type+name filters. Refresh after eval (when chrome **Auto-refresh** is on, default) or the Refresh button. Object-action strip sits at the **bottom** of Objects (more list space): Help · Example · Plot · Names · Structure · **Content** (parent: Head / Tail / Print) · Summary; eraser clears the name regex filter. **Clear** removes visible names from `.GlobalEnv` (confirm). Right-click an object for the Tinn-R-style popup (**Help/Example/Plot/str/ Content/...**).
- **Theme**: Explorer follows Settings theme (whole form + Objects popup glyphs).
- **Companion**: R package `STATghostcom` under `library/STATghostcom_R/STATghostcom` — installed on first use into the active config folder’s `library` (Portable <exe>/data/library or AppData %LOCALAPPDATA%/STATghost/library; see Config); helpers live in exported `.sg` (no attach). Licence **MPL-2.0** (same as STATghost; LICENSE beside the package sources).

9 Settings

- **Appearance** — opacity; **Toolbar** orientation (Separados only — unmarked/disabled under Form único); **Layout** (**Separados** | **Form único**, default Form único; same as Unite/Split — Toolbar and Layout sit side by side); **Icon size** (16 px / 24 px / 32 px, default 24); **Colour theme** — dropdown of all 6 themes (Clas-

sic/Light/ Dark/Contrast/Ocean/Forest) plus a **Next theme (Toggle)** button that cycles them live, one click at a time, wrapping back to Classic after Forest (same live-preview/apply path as the dropdown — Cancel restores the previous theme either way). On Linux (gtk2), native lists, combos, and grid headers follow the STATghost theme (not the desktop GTK theme). Chrome uses flat monochrome glyphs tinted for the active theme (traffic bar, Unidos strip, Plot bar, Explorer). **Package list rows** — **Off** (default) or **Zebra (light)**: alternate-row banding for the Load/Install/Update package grids (Explorer), with a colour-swatch preview next to each option; sits directly below the theme picker.

- **Behavior** — after-eval focus (Auto / Console / Plot / source); Console scroll after eval (Middle / Top / Bottom); Console echo — optional echo of full-line # comments in input chunks (default off); **Config / prefs location** — **Portable** (<exe>/data/, default on a genuine first run of the zip) or **AppData** (%LOCALAPPDATA%/STATghost, existing installs keep this unless switched). Switching copies files across roots (never deletes the originals) and takes effect after restarting STATghost.
- **Session** — Path | Options per language (see Engines); **Ctrl+Tab** / **Shift+Ctrl+Tab** cycle R | Python | Julia pages (does not change Active Use radios).
- **Hotkeys** — OS global: Capture / Arm / Focus / Clear console (**Record** and Clear | Reset; hardened so a stale OS-level grab can never block re-recording the same chord). Capture default stays **Ctrl+C** when Armed; optional alternative chord (Copy in the source app → eval; often the current line with no selection — see Clipboard). **Ctrl+L** clears Console text only (plot unchanged); toolbar Clear still wipes console + plot. While Settings is open, OS global hotkeys are suspended so **Record** can capture the same chords (e.g. Ctrl+L). Alt+Space is rejected (Windows system menu).
- **Shortcuts** — Console-local (not OS RegisterHotKey): **Ctrl+Enter** re-evals the input cell at the caret; **C+Up/C+Down** (hold letter C, not Ctrl) jump previous/next input cell (both configurable).
- **Editor** — font, highlighter, view options (Settings Syntax preview mirrors ATSynEdit Console chrome; live Console uses ATSynEdit + `lexer/`). Under Syntax → Console: chrome **Moldura (frame)** (default) or **Zebrado (zebra)**; optional frame colour, fill, and small corner chamfer. Moldura outlines echoed input only. **Ctrl+Tab** / **Shift+Ctrl+Tab** cycle Font | Syntax.
- **Plot panel** — Display (TImage | AT Gaps), history ←/→ + erase-history, PDF, Size (match pane + DPI).

10 Hotkeys

- **Ctrl+Alt+R** — Arm / Idle (default; Record under Set → Hotkeys).
- **Ctrl+C** — Capture via system clipboard (when Armed).
- Optional Capture alternative (Record under Set → Hotkeys) — Copy in the foreground app, then eval (current line without selection in many editors). Active **only while Armed** (Idle does not register or fire send/eval chords).
- **Ctrl+Alt+F** — Focus mode toggle (default).
- **Ctrl+L** — Clear Console text only (default; R console habit). The Console ATSynEdit keymap does **not** keep the stock Ctrl+L “extend selection by line” binding (would steal Clear).
- **Ctrl+E** — show / hide Explorer (default).
- Brand menu **OS global hotkeys** — when off, Arm / alt-capture / focus / clear / Explorer chords are unregistered so other apps can use them; status shows HK: ON/HK: OFF. Ctrl+C sniper while Armed still works. Opening Settings also suspends those OS chords until the dialog closes (so Record can hear them).

11 Shortcuts (Console)

Local to the Console panel (not OS RegisterHotKey). Set → Shortcuts.

- **Ctrl+Enter** — re-eval the input cell at the caret.

- **C+Up / C+Down** — previous / next input cell (hold letter C, not Ctrl — keeps Ctrl+arrows for scroll).

12 Config

Prefs JSON: <exe>/data/statghost.json (**Portable**, default on a genuine first run of the zip) or %LOCALAPPDATA%\%/STATghost/statghost.json (**AppData**, existing installs, or after switching in Set → Behavior). Package-metadata cache (packages.sqlite3) lives in the same folder.

13 Automated checks (lab)

- **Smoke:** statghost.exe --smoke → statg5_report.txt (RESULT=BOK).
- **Arm on start:** statghost.exe --arm (Armed + session up — handy after a rebuild).
- **SR / SP / SJ / ST:** whole-file samples — --sr (R rterm+rdll), --sp (Python), --sj (Julia), --st / --tf-samples (all) → statg_sr|sp|sj|st_report.txt. Or: powershell -File src/tf_samples.ps1 [-Mode ST|SR|SP|SJ].
- **Dev smokes (SRD|SPD|SJD|STD):** random ~20% of scripts per lane (--srd...--std). Prefer these while iterating.
- **BENCH:** same OLS lesson in all four environments (Rterm.exe | R.dll | Python | Julia) — --bench → w_log/benchmark.txt; powershell -File benchmark/run_bench.ps1.
- **Package dialog probes:** --dbg-pkg-pos drives the real ShowModal for Info/Load/Install/Update across all four Explorer dock modes and captures each dialog's genuine on-screen rect → statg_pkgpos_report.txt; --dbg-pkg-catalog arms a real session, runs a live-CRAN background Refresh, and checks the resulting SQLite cache (row count, empty-title count) → statg_pkgcatalog_report.txt. Both are permanent on-screen regression guards for this class of bug (offline --smoke alone does not exercise real ShowModal placement or a live CRAN fetch).

14 Troubleshooting

- No output — **Armed?** New clipboard text? Correct language under Session? Badge not idle?
- No plot — this chunk must draw (plot, Matplotlib, Plots.jl, ...).
- R.dll won't start — is statghost_rhost.exe next to the app? Is R.dll under R home? Or switch Use to Rterm.exe.
- Python / Julia not found — set path under Session → Path, or install and leave empty for discover.
- Stuck eval — press **Kill**, then Arm again.
- Hotkeys clash with another app — turn off **OS global hotkeys** (sniper Ctrl+C when Armed remains).
- Taskbar icon looks outdated after an update — unpin / repin statghost.exe (Windows caches MAINICON).